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INTERIM FEDERAL SPECIFICATION

PLASTIC STRIP; PLASTIC STRIP WITH ADHESIVE BACKING; SIGN; PLAIN AND REFLECTORIZED

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C., 20406 based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plastic strips, letters, arrows, and special shapes intended for bonding to clean, smooth, dry, non-bleeding asphalt and concrete pavements.

Note: The plastic strips, letters, arrows, and special shapes will be designated traffic markings hereinafter.

1.2 Classification.

1.2.1 Types and classes. The traffic markings shall be of the following types and classes as specified (see 6.2):

Type I - Plain.
Type II - Reflectorized.

Class 1 - With adhesive backing.
Class 2 - Without adhesive backing.

1.2.2 Styles and sizes. The traffic markings shall be of the following styles and sizes as specified (see 6.2):

STYLE A. Arrows

Sizes	
Shank width, inches	Overall length, inches
6	36
12	36
12	72

FSC 9330

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STYLE D. Discs and Squares

Sizes	
Disc diameter, inches	Square (width and length), inches
-	6
12	-
18	-
24	-

STYLE L. Letters, A through Z

Sizes	
Height, inches	Width, inches
12	8
24	12
36	12
48	15

STYLE N. Numerals, 1 through 9 and zero

Sizes	
Height, inches	Width, inches
12	8
24	12
36	12
48	15
72	18

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STYLE S1. Strips, rectangular, cut

Sizes		
Width, inches	Length	
	inches	feet
3	6	-
4	6	-
6	6	-
3	12	-
4	12	-
6	12	-
3	-	20
4	-	20
6	-	20

STYLE S2. Strips, continuous

Sizes	
Width, inches	Length, feet
4	150
6	150

Style X - Special patterns or legends (sizes as specified).

1.2.3 Colors. The traffic markings shall be of the following colors as specified (see 6.2):

White
Yellow (highway)

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-S-300 - Sheeting and Tape, Reflective; Nonexposed Lens, Adhesive Backing.
PPP-B-636 - Box, Fiberboard.

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards:

D 816 - Methods of Testing Rubber Cements.

D 1734 - Making and Preparing Concrete and Masonry Panels for Testing Paint Finishes.

(Copies of the above publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.

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3.1.1 Type I. Type I traffic markings shall consist of not less than 40 percent by weight of flexible vinyl or ethyl cellulose plastic including plasticizers. The remainder of the finished product shall consist of extenders, colorants, and processing aids.

3.1.2 Type II. Type II traffic markings shall consist of not less than 40 percent by weight of flexible vinyl or ethyl cellulose plastic including plasticizers, and not less than 14 percent by weight of clear reflective glass spheres or other reflective materials that satisfy the requirements for reflective intensity specified herein. The reflective glass spheres shall be thoroughly and uniformly dispersed throughout the plastic. At least 80 percent of the reflective glass spheres shall range in size from 0.005 inch to 0.030 inch in diameter and the glass spheres shall contain not more than 20 percent irregular particles or fused spheroids (see 4.4.2). The remainder of the finished product shall consist of extenders, colorants and processing aids.

3.2 Adhesive backing (class 1). Class 1 traffic markings shall have an adhesive backing of pressure-sensitive adhesive or other adhesive system as specified. The adhesive shall be entirely covered with a protective, non-sticky release paper or other release material. The release material shall be easily removed. When tested as specified in 4.4.3, the release material shall be completely removed from the adhesive backing.

3.2.1 Shear strength under static load (class 1). When tested as specified in 4.4.4, the adhesive shall withstand an initial static load of 12.5 pounds per square inch and shall hold for 15 seconds before complete parting of the bond between the adhering surfaces.

3.3 Thickness. The thickness of the traffic markings shall be not less than that specified in the contract or order (see 6.2). The thickness shall not exceed that specified by more than 20 percent (see 4.4.5).

3.4 Sizes. Traffic markings shall have the dimensions as specified in the contract or order (see 1.2.2 and 6.2). Dimensional tolerances shall be as specified in table I (see 4.4.6).

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TABLE I. Tolerances for dimensions

	Tolerance, plus and minus												
	Inches											Feet	
	3	4	6	8	12	15	18	24	36	48	72	20	150
						STRIPS							
Width	1/32	1/16	3/32										
Length			3/32		3/32							1/4	1/2
						DISCS							
Diameter					1/16		1/16	1/16					
						SQUARES							
Width and Length			3/32										
						ARROWS							
Shank width			3/32		3/32								
Length(overall)									3/32		1/8		
						LETTERS							
Width a/				3/32	3/32	3/32							
Height a/					3/32			3/32	3/32	1/8			
						NUMERALS							
Width a/				3/32	3/32	3/32	3/32						
Height a/					3/32			3/32	3/32	1/8	1/8		

a/ Measured between outermost points on a straight line, unless otherwise specified.

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3.5 Color. Traffic markings shall be pigmented white or yellow (highway) as specified in the contract or order (see 1.2.3 and 6.2). Titanium dioxide shall constitute the white pigment, and chrome yellow (medium), the yellow pigment. Coloring shall be constant throughout the cross section of the traffic markings (see 4.4.7).

3.6 Wear resistance. When tested as specified in 4.4.8, the rate of wear shall not exceed 0.5 gram per 1,000 revolutions of the abrasion tester.

3.7 Tensile strength and elongation. When tested as specified in 4.4.9, the traffic markings of vinyl plastic shall have a minimum tensile strength of 600 pounds per square inch and a maximum elongation of 70 percent; traffic markings of ethyl cellulose plastic shall have a minimum tensile strength of 250 pounds per square inch and a minimum elongation of 50 percent.

3.8 Cold crack resistance. When the vinyl plastic traffic markings are exposed to 32° F., or 50° F. in the case of ethyl cellulose plastic traffic markings, and then bent as specified in 4.4.10, the material shall not break, chip, or crack.

3.9 Reflective intensity (type II). When tested as specified in 4.4.11, the reflective intensity of the type II material shall be greater than 0.1 at 0.2 degree divergence angle and 85 degrees incidence angle.

3.10 Environmental resistance.

3.10.1 Accelerated weathering. After 120 hours exposure to light and water as specified in 4.4.12, the traffic markings shall show no evidence of cracking, blistering or discoloration. The ethyl cellulose material will show some yellowing after this exposure which is to be tolerated.

3.10.2 Chemical resistance. Traffic markings shall not disintegrate when in contact with each of the following agents: gasoline, motor oil (10-30W), and 10 percent calcium chloride solution (see 4.4.13).

3.11 Installation instructions. The supplier shall furnish detailed instructions on the recommended methods of installing the traffic markings. This information shall be provided in printed form and shall be attached to the material.

3.12 Workmanship. Traffic markings shall be free of ragged edges, die lines, cracks, holes, pits, blisters, discolorations and other defects which would affect the serviceability of the material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be made according to MIL-STD-105. An inspection lot shall consist of all material of the same type, class, style and size submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected as required by the referenced specifications to the extent specified.

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4.2.2 Inspection of traffic markings.

4.2.2.1 Examination of traffic markings. Examination shall be made according to the classification of defects, inspection levels, and acceptable quality levels (AQLs) specified in 4.2.2.1.4. The sample size shall be expressed in units of individual cut pieces or rolls, as applicable, for the examination in 4.2.2.1.1 and 4.2.2.1.2, and in units of shipping containers for the examination in 4.2.2.1.3

4.2.2.1.1 Examination of the traffic markings for defects in appearance and workmanship. The sample unit for the examination specified in table II shall be one cut piece or three linear feet of roll material.

TABLE II. Examination of the traffic markings for defects in appearance and workmanship

Examine	Defect
Appearance	Color not as specified. Type, class, and style not as specified.
Workmanship	Any ragged edges, cracks, holes, blisters, and dirt.

4.2.2.1.2 Examination of the traffic markings for dimensional defects. The sample unit for the examination specified in table III shall be one cut piece or three linear feet of roll material.

TABLE III. Examination of the traffic markings for dimensional defects

Examine	Defect
Width	Varies by more than tolerance specified (see table I); not according to contract requirements.
Length	Varies by more than tolerance specified (see table I); not according to contract requirements.
Diameter	Varies by more than tolerance specified (see table I); not according to contract requirements.

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TABLE III. Examination of the traffic markings for dimensional defects(continued)

Examine	Defect
Height	Varies by more than tolerance specified (see table I); not according to contract requirements.
Thickness	Varies by more than tolerance specified (see 3.3); not according to contract requirements.

4.2.2.1.3 Examination of preparation for delivery. An examination shall be made in accordance with table IV to determine that the packing, and marking comply with the requirements in 5.1 and 5.2. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to closing.

TABLE IV. Examination of preparation for delivery

Examine	Defect
Packing	Not level specified; not according to contract requirements.
Marking	Marking omitted, illegible, incorrect, incomplete, or not according to contract requirements.
Weight	Gross exceeds specified requirements

4.2.2.1.4 Inspection levels and AQLs (AQLs for examination). The inspection levels for determining the sample size of AQLs expressed in defects per hundred units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	I	2.5
4.2.2.1.2	S-2	2.5
4.2.2.1.3	S-2	4.0

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4.3 Lot acceptance testing. The traffic markings shall be tested for the properties specified in section 3. The sample unit shall be 6 cut pieces or 6 square feet of roll material selected at random. The inspection level for determining the sample size for test shall be level S-1. The lot size shall be expressed in units of cut pieces or rolls, as applicable.

4.4 Test methods.

4.4.1 Conditioning. Unless otherwise specified, test specimens shall be conditioned as specified in Fed. Test Method Std. No. 406 and tested under these conditions.

4.4.2 Glass sphere test (by extraction). Cut a specimen into 1/4 inch pieces and immerse the pieces in lacquer thinner warmed to 130° F. Stir the mixture until the plastic dissolves and the spheres settle in the beaker. Decant the mixture and repeat the process until the spheres are clean. Dry the spheres and them compute the weight percent.

4.4.3 Release material removal. Cut a specimen, 1/2 inch by 6 inch, of the class I traffic marking. Detach the release material for one inch of the length and clamp the specimen (face inward) to a vertical surface at the point where the release material was removed. To the end of the partly removed release material, clamp a supported one pound weight. Free the one pound load and let fall. Examine the specimen for unremoved release material.

4.4.4 Shear strength under static load. Cut a class 1 adhesive coated panel 3 inches wide by 6 inches long. Remove the release material completely. Adhere the panel to a cement block prepared according to ASTM D 1734, and set the panel for a one (1) inch overlap of its length. A pressure of 50 pounds per square inch shall be applied for a period of thirty (30) seconds to the sample. An initial, momentary application of thirty-seven and one half (37.5) pounds of load shall produce no slippage when applied to the bonded specimen vertically suspended with full width clamps, as described in ASTM D 816. After the momentary application, the 37.5-pound test load shall be hung on the lower clamp and this static shear load shall not cause complete separation before 15 seconds have elapsed.

4.4.5 Thickness. Thickness measurements shall be made with a calibrated micrometer or accurate thickness gauge capable of measuring to 0.001 inch.

4.4.6 Length and width. The length and width of the sheet shall be measured with a calibrated steel rule or tape.

4.4.7 Color. Determine compliance with 3.5 by accepted chemical identity and visual methods.

4.4.8 Wear resistance. Determine the wear resistance in accordance with method 1091 of Fed. Test. Method Std. No. 406 except that H-18 Calabrade wheels shall be used and the load on each wheel shall be 1000 grams. The test specimen shall be subjected to an abrasion of 2500 cycles.

4.4.9 Tensile strength and elongation. Determine the tensile strength and elongation of the material in accordance with method 1011 of Fed. Test Method Std. No. 406. Speed of testing shall be 0.25 inches per minute.

4.4.10 Cold crack resistance. Immerse the vinyl plastic in an ice bath for two (2) hours. In the case of ethyl cellulose plastic, immerse in a water bath at 50° F. for two (2) hours. Then remove the specimen and bend abruptly into a right angle. Examine the specimen for cracks, chips, or breaks.

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4.4.11 Reflective intensity. Determine the reflective intensity in accordance with the procedures given in L-S-300.

4.4.12 Accelerated weathering. Specimens shall be subjected to accelerated weathering according to method 6022 of Fed. Test Method Std. No. 406.

4.4.13 Chemical resistance. Cut three specimens, 3 inches by 3 inches. With a pipette, apply one milliliter of each reagent specified in 3.10.2 to the center of each specimen held in a level position. Reagents shall remain in contact with the specimens for four (4) hours in an atmosphere at 23° ± 1.1° C. (73.5° ± 2° F.)(Exercise precaution in the handling of gasoline). At the end of this period, wash off the reagent and examine specimens for any effects of deterioration.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Rolls. The material furnished in continuous lengths of 20 or 150 feet shall be packed in center mandrel cartons which will permit dispensing of the material through a slot provided in the end of the carton.

5.1.1.2 Cut pieces. The material furnished in cut pieces shall be packed in overseas class or type fiberboard boxes conforming to PPP-B-636. The gross weight shall not exceed the weight limitation of the box specification. Closure and strapping shall be in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Rolls. The material furnished in continuous lengths of 20 or 150 feet shall be packed in center mandrel cartons which will permit dispensing of the material through a slot provided in the end of the carton.

5.1.2.2 Cut pieces. The material furnished in cut pieces shall be packed in domestic class or type fiberboard boxes conforming to PPP-B-636. The gross weight shall not exceed the weight limitation of the box specification. Closures and strapping shall be in accordance with the appendix of the box specification.

5.1.3 Level C.

5.1.3.1 Rolls and cut pieces. The material shall be packed to afford adequate protection against damage during shipment within the continental United States direct from the supply source to the receiving activity. This level, as a minimum, will conform to applicable carrier rules and regulations.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

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6.1 Intended use. The Type I traffic markings are intended for application to asphalt and concrete pavements where a nonreflective surface is required. Type II is retroreflective.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and, date of this specification.
(b) Type, class, style and size required (see 1.2).
(c) Thickness required (see 3.3).
(d) Color desired (see 1.2.3).
(e) Kind of adhesive system desired (see 3.2).
(f) Selection of applicable level of packing required (see 5.1).

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